



Eric Araújo

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Welcome

I investigate the spread of opinions, sentiments, and behaviors in complex systems through multiagent modeling and simulations, offering insights into social dynamics in politics, cooperation, health, criminology, and religion.

The work I develop is multidisciplinary, using agent-based models and data science to solve problems in areas such as political polarization in churches, public security, the spread of fake news, and promoting healthier lifestyles.

Education

- Ph.D. Vrije Universiteit Amsterdam**, Computer Science Amsterdam, NL
• 2026 Nagel Institute Fellowship Aug 2014 – Sept 2018
• 2026 McGregor Summer Research Fellowship
• 2026/2027 Calvin Research Fellowship
• Thesis: [Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks](#) [↗](#)
- M.Sc. Universidade Federal de Minas Gerais**, Computer Science Belo Horizonte, Brazil
Mar 2007 – Sept 2009
- B.Sc. Universidade Federal de Viçosa**, Computer Science Viçosa, Brazil
Mar 2003 – Dec 2006

Experience

- Calvin University**, Associate Professor Grand Rapids MI USA
• Computer Science Department Calvin University July 2024 – present
- FIOCRUZ**, Researcher Rio de Janeiro Brazil
• FIOCRUZ is a leading institution in health research and development in Brazil. My work focused on collecting data and analyzing it to understand medical circulation of health sciences among Brazilian researchers. 2022 – 2024
- Vrije Universiteit Amsterdam**, Guest Researcher Oct 2018 – Apr 2022
• Collaborator in the Department of Computer Science Vrije Universiteit Amsterdam
• Worked on criminology models with Prof. Dr. Charlotte Gerritsen.
- Universidade Federal de Lavras**, Associate Professor Minas Gerais Brazil
• Computer Science Department Universidade Federal de Lavras Feb 2011 – June 2024
- Gammon Presbyterian Faculty**, Professor Minas Gerais Brazil
• Professor and Coordinator of the Information Systems course Jan 2010 – Jan 2011

Projects

- Procores** [↗](#) UFMG Brazil
Research group funded by CNPq (Brazil) May 2022 – present

- ProCoReS is a research group funded by CNPq, dedicated to the study of social networks, mathematical modeling, contagion processes, and the analysis of complex data. The group brings together researchers from different fields to propose and analyze innovative models that consider multiple aspects of social networks and their dynamics.
- To model networks and contagion, most current approaches use graphs. However, these are not entirely realistic, and new models of contagion processes should allow for: representation of different classes of individuals and connections, linkage between concurrent and co-occurring processes, and connections involving two or more individuals simultaneously. Understanding how connections are formed and how different types of flows permeate such networks is crucial to understanding how our society is organized and evolves. The goal is to propose mathematical, statistical, agent-based, and machine learning models that account for: multiple complex aspects of social contagion processes and parameterization through data sources (e.g., online social networks, mobility, collaborations). Such models should allow for classes of individuals and relationships; model co-occurrence, causality, and links between contagion processes; and represent multi-relationships.
- Funded by: CNPq Universal 2022

Socioeconomic Impacts in Territories Affected by Technological Disasters

- This project investigates the socioeconomic consequences of technological disasters in Brazil, which have increased significantly in recent decades. Between 2000 and 2020, Brazil recorded 96 technological disasters—ranging from dam collapses to oil spills—impacting over 42 million people and causing over USD 17 billion in damages (EM-DAT, 2021). Unlike natural disasters, which occur without human interference, technological disasters result from anthropogenic actions, such as industrial accidents or environmental contamination. These events often lead to material loss, environmental degradation, and serious public health issues, demanding robust mechanisms for community response and risk management.
- By analyzing historical data and specific case studies such as the Mariana (2015) and Brumadinho (2019) dam collapses, the project seeks to understand how these disasters affect the social and economic fabric of impacted territories. The ultimate goal is to identify patterns and develop frameworks that can inform policies for disaster prevention, mitigation, and recovery—contributing to more resilient communities in the face of growing technological risks.

UFLA Brazil
Sept 2022 – Feb 2024

Analysis of User Behavior on a University Campus Wi-Fi Network

Masters Project by Thiago do Prado Ramos

- This project aims to analyze the behavior of users on a university campus Wi-Fi network, focusing on connection patterns and user mobility. By examining connection logs, the project seeks to understand how users interact with the network, their mobility patterns, and how these factors influence network performance and user experience.
- The findings will help in optimizing network infrastructure, improving user experience, and informing future network design and management strategies.

UFLA Brazil
Mar 2021 – Sept 2023

Algorithms for the Optimization of Supply Chain Management Systems

Project in partnership with Prof. Mayron Moreira (UFLA Brazil)

- This project developed and evaluated algorithms aimed at optimizing key components of supply chain management, including inventory control, logistics, and production scheduling. By modeling supply chains as dynamic systems, the project applied heuristic and metaheuristic approaches—such as genetic algorithms and simulated annealing—to improve efficiency, reduce costs, and enhance decision-making under uncertainty.

UFLA Brazil
Mar 2021 – Mar 2022

- The research focused on real-world constraints such as demand variability, delivery time windows, and resource limitations. The resulting algorithms demonstrated significant improvements in performance when compared to traditional methods, and provided a flexible framework that can be adapted to various industrial sectors seeking to enhance their operational processes.

Urban Tree Cover and Crime

Project in partnership with Prof. Michele Valquíria dos Reis (UFLA Brazil)

UFLA Brazil

Mar 2020 – Mar 2024

- Urban tree cover provides vital ecosystem services that promote public health and well-being. However, there is a common perception—found in media reports and informal discussions—that trees may facilitate crime by obstructing visibility or serving as hiding places. While some studies have linked vegetation to fear of crime, often recommending vegetation removal to improve safety, research from the Northern Hemisphere increasingly suggests a negative correlation between urban greenery and crime rates, indicating that vegetation may actually contribute to crime reduction.
- This project aims to investigate the relationship between urban tree cover and crime incidence in Brazilian cities. It integrates data from street tree inventories, georeferenced crime records from the Military Police, socioeconomic indicators from the national census (IBGE), citizen surveys on perceptions of vegetation and safety, and citizen requests for tree removal. The goal is to generate scientific insights into the social dimensions of urban greenery and to inform evidence-based environmental management policies. The findings are intended to support public agencies and civil society in designing safer, greener urban spaces.

Computational Modeling and Simulation of Contagion in Social Networks

UFLA Brazil

Sept 2018 – June 2024

- Social contagion refers to the spread of behaviors, perceptions, and emotions through interactions among individuals. Often occurring unconsciously, this phenomenon plays a role in shaping collective opinion, promoting positive behaviors, and influencing public discourse. This project aims to simulate the diffusion of information originating from social media platforms across various contexts, using agent-based and network modeling approaches.
- The study investigates how network topology, individual agent characteristics, and intervention strategies affect the spread and quality of information. It explores methods for enhancing the visibility and influence of high-quality content while limiting the reach of misinformation or fake news. The results aim to inform the design of healthier and more reliable information ecosystems, with potential applications in communication policy, platform design, and digital education.

Research Skills

1. **Agent-based modeling:** Netlogo and Python tools for ABMs.
2. **Data Science and Machine Learning:** Python, R, Statistics.
3. **Social Network Analysis:** Gephi, NetworkX.
4. **Network-oriented modeling:** Techniques for modeling complex systems using network-oriented methods.
5. **Programming:** Proficient with Python, C++, Netlogo, and Git; good understanding of Web development.
6. **Languages:** English (fluent), Portuguese (native), Dutch (basic).

Current Teaching

CS112 - Introduction to Data Structures [↗](#)

Calvin University USA

An introduction to data structures and algorithms, focusing on the design and analysis of algorithms, data structures, and their applications.

Sept 2024 – present

CS354A - Databases Management Systems [↗](#)

Calvin University USA

An introduction to database management systems, covering data modeling, query languages, and database design.	Jan 2025 – present
CS300b - Agent-based Modeling An introduction to agent-based modeling, focusing on the design and implementation of agent-based models for complex systems.	Calvin University USA Sept 2024 – present
CS214 - Concepts of Programming Languages ↗ Concepts and paradigms of programming languages.	Calvin University USA Jan 2026 – present

Recent Events

NetLogo Conference 2026 ↗ Supervised student presentation: Katelin Jandris, Ryan Klein, and Ovgu Tufan presented <i>Cognitive Modeling of Church Polarization</i> (senior project).	Chicago IL USA July 2026 – July 2026
2026 Technology and the Human Person in the Age of AI ↗ Presented my framework of Behavior, Belonging, and Beliefs during the event. The title of the presentation was ‘Cognitive Modeling of Christian Communities - Developing Intelligent Agents to Address the Polarization Effects in Churches’	Waco TX USA Feb 2026 – Feb 2026
Grace Hopper Celebration 2025 ↗ The Grace Hopper Celebration (GHC) is the worlds largest gathering of women in computing. I am the main advisor of the Girls Who Code group at Calvin University.	Chicago IL USA Nov 2025 – Nov 2025
ASA 2025 Annual Meeting ↗ Science & Technology as Worship	Lakewood CO USA July 2025 – July 2025
Autonomous Agents for Social Good 2025 ↗ 6th International Workshop on Autonomous Agents for Social Good (AASG) 2025 in conjunction with the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025).	Detroit MI USA May 2025 – present

Publications

From brain drain to brain circulation: International researcher mobility in Brazilian health and life sciences (2005–2020) Biazoli, Leonardo, Albuquerque, Priscila Costa, Araújo, Eric, de Oliveira, Izabela Regina Cardoso, Fonseca, Bruna de Paula 10.1162/QSS.a.411 ↗ (Quantitative Science Studies)	Feb 2026
Exploring the link between urban topology and street crime using complex networks: a case study from Southeast Brazil Matheus de Andrade Flausino, Eric Araújo , Angélica Sousa da Mata 10.1093/comnet/cnaf016 ↗ (Journal of Complex Networks)	July 2025
The Use of Agent-based Modeling in the Study of Complex Systems (In Portuguese) Clayton R. da Silva, Olivia Mesquita, Eric Araújo , Angélica S. Mata 10.1590/1806-9126-RBEF-2024-0464 ↗ (Revista Brasileira de Ensino de Física)	2025
Understanding the complexities of Bluetooth for representing real-life social networks Simoski, Bojan, Klein, Michel C.A., Araújo, Eric Fernandes de Mello, van Halteren, Aart T., van Woudenberg, Thabo J., Bevelander, Kirsten E., Buijzen, Moniek, Bal, Henri 10.1007/s00779-020-01435-x ↗ (Personal and Ubiquitous Computing)	Feb 2024
Police Planning Based on the Use of Business Intelligence (BI): The Police Manager’s Perception Regarding the Violent Crime Rate in the 8th Battalion of the Military Police of Minas Gerais (PMMG) (In Portuguese) Erich da Costa, Denis Renato de Oliveira, Eric Araújo	2023

[10.29377/rebsp.v16i01.614](#) [↗](#) (Revista Brasileira de Estudos de Segurança Pública)

Epidemiological Intelligence in Operation COVID-19: A New Decision Support Tool in Brazil's Military Logistics (*In Portuguese*) 2023

José Roberto Pinho de Andrade Lima, Mariza Ferro, **Eric Araújo**, Cristiano Barros, Ernesto Rademaker Martins, Beatriz Helena Felício Fuck Telles Ferreira

[10.47240/revistadaesg.v38i82.1250](#) [↗](#) (Revista da Escola Superior de Guerra)

Political Polarization on Twitter During the COVID-19 Pandemic: A Case Study in Brazil 2022

Pedro Brum, Matheus Cândido Teixeira, Renato Vimieiro, **Eric Araújo**, Wagner Meira Jr, Gisele Lobo Pappa

[10.1007/s13278-022-00949-x](#) [↗](#) (Social Network Analysis and Mining)

Urban Afforestation and Public Safety: A Bibliometric Study Using the Consolidated Meta-Analytical Focus Theory (*In Portuguese*) 2022

Kelly Iapuque Rodrigues de Sousa, Michele Valquíria dos Reis, Rafael Rodrigues de Castro, **Eric Araújo**

[10.5585/geas.v11i2.22965](#) [↗](#) (Revista de Gestão Ambiental e Sustentabilidade (GeAS))

Detecção e Classificação de Bots Utilizando Redes Neurais Artificiais e Análise de Sentimentos 2021

Gabrieli Silva, Eliaquim Ramos, Eric Araújo, Fábio Borges, Mariza Ferro

[www.eamc.lncc.br/proceedings.html](#) [↗](#) (Proceedings of the XIV Encontro Acadêmico de Modelagem Computacional)

The Virus and Socioeconomic Inequality: An Agent-based Model to Simulate and Assess the Impact of Interventions to Reduce the Spread of COVID-19 in Rio de Janeiro, Brazil Apr 2020

Vinícius Prata Klôh, Gabrieli Dutra Silva, Mariza Ferro, **Eric Araújo**, Cristiano Barros de Melo, José Roberto Pinho de Andrade Lima, Ernesto Rademaker Martins

[10.34119/bjhrv3n2-192](#) [↗](#) (Brazilian Journal of Health Review)

Disconnecting for the Good: A Network-Oriented Model for Social Contagion of Opinions and Social Network Interventions to Increase Adherence to Social Distancing 2020

Eric Araújo, Mariza Ferro, Gabrieli Silva

[10.5753/brasnam.2020.11170](#) [↗](#) (Proceedings of the IX Brazilian Workshop on Social Network Analysis and Mining)

Creating a Temporal Pattern for Street Robberies Using ABM and Data from a Small City in Southeast Brazil 2020

Eric Araújo, Charlotte Gerritsen

[10.4324/9780429277177](#) [↗](#) (Agent-Based Modelling for Criminological Theory Testing and Development)

Identifying Influence Agents That Promote Physical Activity Through the Simulation of Social Network Interventions: Agent-Based Modeling Study Aug 2019

Thabo J. van Woudenberg, Bojan Simoski, **Eric Araújo**, Kirsten E. Bevelander, William J. Burk, Crystal R. Smit, Laura Buijs, Michel Klein, Moniek Buijzen

[10.2196/12914](#) [↗](#) (Journal of Medical Internet Research)

A Social Network Model for Integration of Refugees 2019

Fabio Curi, Dimitris Nikolopoulos, **Eric Araújo**

[10.5220/0007930601650175](#) [↗](#) (Proceedings of the 9th International Conference on Simulation and Modeling Methodologies, Technologies and Applications)

Parameter Optimization for Deriving Bluetooth-Based Social Network Graphs 2019

Bojan Simoski, Michel C.A. Klein, **Eric Araújo**, Aart T. Van Halteren, Thabo Van Woudenberg, Kirsten E. Bevelander, Moniek Buijzen, Henri Bal

[10.1109/SmartWorld-UIC-ATC-SCALCOM-IOP-SCI.2019.00318](#) [↗](#) (2019 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI))

- Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks (PhD Thesis)** 2018
Eric Araújo
research.vu.nl/en/publications/632878cc-e2e7-40ab-a34e-91bde40eaf50
- Using Simulations for Exploring Interventions in Social Networks - Modeling Physical Activity Behaviour in Dutch School Classes** 2018
Eric Araújo, Bojan Simoski, Thabo van Woudenberg, Kirsten Bevelander, Crystal Smit, Laura Buijs, Michel Klein, Moniek Buijzen
[10.5220/0006857704140425](https://doi.org/10.5220/0006857704140425) (Proceedings of 8th International Conference on Simulation and Modeling Methodologies, Technologies and Applications - SIMULTECH)
- Applying Machine Learning Algorithms for Deriving Personality Traits in Social Network** 2018
Eric Araújo, Bojan Simoski, Michel Klein
[10.1145/3167132.3167377](https://doi.org/10.1145/3167132.3167377) (Proceedings of the 33rd Annual ACM Symposium on Applied Computing)
- Detecting Dutch Political Tweets: A Classifier based on Voting System using Supervised Learning** 2018
Eric Araújo, Dave Ebbelaar
[10.5220/0006592004620469](https://doi.org/10.5220/0006592004620469) (Proceedings of the 10th International Conference on Agents and Artificial Intelligence - Volume 1: ICAART)
- Social Connection Dynamics in a Health Promotion Network** 2017
Eric Araújo, Michel Klein, Aart van Halteren
[10.1007/978-3-319-50901-3_66](https://doi.org/10.1007/978-3-319-50901-3_66) (Complex Networks & Their Applications V)
- Explaining Changes in Physical Activity Through a Computational Model of Social Contagion** 2017
Julia S. Mollee, **Eric Araújo**, Adnan Manzoor, Aart T. van Halteren, Michel Klein
[10.1007/978-3-319-54241-6_19](https://doi.org/10.1007/978-3-319-54241-6_19) (Complex Networks VIII)
- Exploring Parameter Tuning for Analysis and Optimization of a Computational Model** 2017
Julia S. Mollee, **Eric Araújo**, Michel C. A. Klein
[10.1007/978-3-319-60045-1_36](https://doi.org/10.1007/978-3-319-60045-1_36) (Advances in Artificial Intelligence: From Theory to Practice)
- A Temporal-Causal Model for Spread of Messages in Disasters** 2017
Eric Araújo, Annelore Franke, Rukshar Wagid Hosain
[10.1007/978-3-319-67077-5_37](https://doi.org/10.1007/978-3-319-67077-5_37) (Computational Collective Intelligence)
- A Computational Cognitive Model for Political Positioning and Reactions in Web Media** 2017
Eric Araújo, Michel Klein
[10.1109/ICCI-CC.2017.8109782](https://doi.org/10.1109/ICCI-CC.2017.8109782) (2017 IEEE 16th International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC))
- Analysis and Refinement of a Temporal-Causal Network Model for Absorption of Emotions** 2016
Eric Araújo, Jan Treur
[10.1007/978-3-319-45243-2_3](https://doi.org/10.1007/978-3-319-45243-2_3) (Computational Collective Intelligence)
- Online Sharing of Physical Activity: Does It Accelerate the Impact of a Health Promotion Program?** 2016
Adnan Manzoor, Julia S. Mollee, **Eric Araújo**, Aart T. van Halteren, Michel C. A. Klein

[10.1109/BDCloud-SocialCom-SustainCom.2016.40](#) [↗](#) (2016 IEEE International Conferences on Big Data and Cloud Computing (BDCloud), Social Computing and Networking (SocialCom), Sustainable Computing and Communications (SustainCom) (BDCloud-SocialCom-SustainCom))

Analysis and Evaluation of Social Contagion of Physical Activity in a Group of Young Adults 2015

Eric Araújo, Anita V. T. T. Tran, Julia S. Mollee, Michel C. A. Klein

[10.1145/2818869.2818922](#) [↗](#) (Proceedings of the ASE BigData & SocialInformatics 2015)

An Analysis of TCP Protocol Issues in Asymmetric Networks and Existing Solutions (In Portuguese) - M.Sc. Thesis 2009

Eric Araújo

hdl.handle.net/1843/BUBD-9JWQ36 [↗](#)

AnimEasy: Free Tool for Digital Inclusion Focused on Multimedia Using Inkscape (In Portuguese) - B.Sc. Senior Project 2007

Eric Araújo, Ricardo dos Santos Ferreira

VII Simpósio de Informática do Planalto Médio (SIPM 2007)